PSeq Technical Brief

Full-length, isoform-resolved RNA transcriptomes using NGS ● ML-ready outputs

Full-length RNA sequencing on short-read NGS sequencers

Profiling full-length RNA without third-generation instruments (such as Pacific Biosciences or Oxford Nanopore) is highly valuable, primarily because it enables the precise identification of complex RNA isoforms, alternative splicing, and novel transcripts using the high throughput, high accuracy and low cost of short-read technologies (e.g., Illumina).

Quantitative isoform sequencing for all genes, at scale

PSeq resolves full-length *individual* molecules – ***not merely 'kinds' of molecules*** – at a scale approaching solution chemistry (>*fmol*). PSeq resolves the full diversity of alternatively spliced, alternatively edited isoforms, including products of gene fusion, and the full range of lncRNAs. PSeq enables sequencing of isoform libraries of individual genes or multigene families.

Technically, large-scale single-molecule resolution is a revolutionary advance in fundamental chemistry.

Industry-leading accuracy:

The best accuracy currently available for Third-generation platforms – 99.9% – means that an 18 kb Nipah virus genome will average 18 sequencing errors per molecule. Phaeno NGS PSeq can sequence cDNAs essentially error free, exceeding the accuracy of enzymatic reverse transcription (~ 10^{-5} errors per base – a 100-fold improvement).

NGS instrument (Illumina) base-calling accuracy ranges between 99.9% - 99.99% (Q30-40). PSeq sequences each RNA base at 5-100x for **consensus accuracy of <1 error per 10^{15} bases**. This translates into sequencing > 10^{11} human RNAs before the first platform generated sequencing error, exceeding the accuracy of enzymatic reverse transcription).

Automated data pipeline

Unlike RNA-Seq, PSeq data assembly does not rely on statistical algorithms or prior information. Unlike Third Generation technology, computation is fully automated. PSeq data assembly is rapid and highly scalable for sample size, and all information from raw FASTA files through assembled molecules are accessible for in depth validation. Data assembly for 10M RNAs or 500M RNAs can be performed in a few hours on appropriate EC2 AWS instances.

Data and delivery

Information for each molecule, and the compiled run, will be provided for client upload: data can be assembled in any configuration, or provided for upload to client data processors for interoperability with tools for data retrieval, quantification, visualization or browsing.

Information accessible or directly computable for each molecule encompasses:

- individual FASTA files for each library fragment paired-end sequence in the sequencing run
- assembled sequences (UTRs, ORF, exon junctions, predicted AA sequence and molecular weight).
- updated annotation of discovered exons and 5' and 3' boundaries of each source gene sequence.

Customer data can be securely downloaded from our customer portal or via SFTP.

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Library chemistry

Phaeno chemistry generates paired-end DNA sequencing libraries compatible with any unmodified NGS platform. Each source RNA molecule is labeled with a Source Molecule Identifier (SMID) comprising 70B–4.3T unique barcodes, multiplex markers, and strand-sense information. Library preparation replicates each tagged cDNA without PCR, using intramolecular rearrangements to produce random internal fragments in which the SMID is encoded at the start of Read 1 in an otherwise conventional paired-end library.

Data processing

The proprietary automated data pipeline is uploaded with FASTA files for each library fragment derived from paired-end NGS. Read pairs are:

- Sorted according to (SMID) barcodes.
- Trimmed to remove tags
- Assembled by overlap as a continuous single-molecule transcript.
- Source gene Identified and bounded sequences retrieved from human genome database:
- Structural information assembled for each RNA for visualization, transcriptome compilation and analysis.

Security, data storage and retention

Phaeno's security, data storage and retention policies are available separately upon request.

Cloning

Tools will be available to clone ***individual cDNA molecules*** discovered during sequencing:

bio-bank DNA → generate RNA → crystallography grade protein

Key metrics

Phaeno library chemistry reliably preserves individual source-molecule identity across library preparation, sequencing, and data assembly. SMID barcodes are efficiently transferred to each read fragment and positioned at the start of Read 1. Data-pipeline analysis shows negligible barcode chimerism, enabling accurate detection of gene fusions and RNA virus whole-genome sequencing. PSeq further enables a novel RNA-sequencing metric: the maximum source-molecule length over which a defined number of molecules can be sequenced before platform-generated errors occur. As an emerging technology, some capabilities are not yet RUO-commercialized, and Phaeno welcomes collaboration with academic, commercial, and public-health KOLs to address specific technical needs.

phaeno
5270 California Avenue • Suite 300 • Irvine, CA 92617
info@phaenobiotech.com
www.phaenobiotech.com

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PSEQ VERSUS RNA-SEQ AND 3RD GENERATION TECHNOLOGY

Metric	PSEQ	RNA-Seq [1]	3rd-Gen Tools [2]
Qualitative Metrics			
Data Quality	✓	✗ Systematic end bias; incomplete transcript coverage	✗ Consensus polishing required; rare variants often lost
Isoform resolution	✓	✗	✓
Fully assembled transcripts	✓ Every molecule is assembled (5' to 3').	✗ Fragments (50-300 bp) require statistical reconstruction.	✓
Direct cloning	✓	✗	✗
Quantitative	✓ Fully-quantitative (molecule -counting, no inference)	✗ Semi-quantitative	✗ Semi-quantitative
Simple Workflow	✓	✓ (but multi-step & model-dependent)	✗
Throughput	✓ High	✓ High	✗ Low
Full UTR resolution	✓	✗	✗ Partial Only
NGS Compatible	✓	✓	✗ Requires proprietary 3rd-generation equipment.
AI/ML Optimized	✓	✗	✗
Single-cell resolution	✓	✓ With add-on chemistry	✗
Performance Characteristics[3]			
Rare[4] isoform sensitivity	**>87%**	<5%	~45%
Full-length isoform recovery rate	**>94%**	~72%	>88%
Median transcript length resolved	3.2kb (up to 12kb)	1.1kb (fragmented)	2.8kb
Hands-on time per sample	<2 hours	~3 hours	~6 hours
Turn-around time (library to FASTA)	<24 hours	~26 hours	~72 hours

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[1] Illumina TruSeq Stranded mRNA, STAR + RSEM quantification (v2.7).
[2] PacBio Iso-Seq3, SMRT-Link v11, consensus polishing.
[3] PSeq performance metrics are preliminary and may be refined as additional samples are evaluated under increasingly stringent validation conditions.
[4] < 0.01% abundance.